Frequency-Enhanced Spatial–Spectral Network for Hyperspectral Imagery Reconstruction From Multispectral Imagery

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***Abstract*—Hyperspectral imagery (HSI) delivers detailed spectral information critical for remote sensing applications such as high-accuracy land-cover classification, quantitative parameter retrieval, and environmental monitoring. However, satellite-borne HSI often suffers from limited spatial resolution owing to inherent sensor constraints, whereas airborne HSI is constrained by restricted spatial coverage. Reconstructing high-resolution HSI from multispectral imagery (MSI) emerges as a promising strategy to address these challenges. In this study, we propose the frequency-enhanced spatial–spectral network (FESSN), a computationally efficient architecture that innovatively integrates multidomain fusion across spatial, spectral, and frequency domains to achieve superior reconstruction performance. A key innovation is the neural network-driven frequency-enhanced modulation (FEM), which adaptively refines spectral amplitudes and phases via fast Fourier transform (FFT), providing interpretable, parameter-efficient enhancements to bridge spatial–spectral modeling gaps. Meanwhile, a Mamba-based multiscale spatial fusion module (MMSAF) that seamlessly integrates local features with long-range dependencies, and a U-shaped spectral module (USEM) that integrates Mamba and attention mechanisms to model intergroup and intragroup relationships, while adhering to spectral sparsity priors. The experimental results demonstrate FESSN outperforming six state-of-the-art methods in metrics like root-mean-squared error (RMSE) (up to 5.31% improvement), peak signal-to-noise ratio (PSNR), spectral angle mapper (SAM), relative global dimensional error in synthesis (ERGAS), and structural similarity index measure (SSIM). Downstream tasks in land-cover classification further validate its utility, positioning FESSN as a breakthrough in efficient, high-fidelity HSI reconstruction. To facilitate reproducibility and further research, the code will be publicly available at: https://github.com/KustAIRS/TGRS-FESSN**

***Index Terms*—Attention mechanisms, fast Fourier transform (FFT), Mamba, remote sensing, spectral reconstruction.**

I. Introduction

HYPERSPECTRAL remote sensing acquires continuous spectral signatures across hundreds of bands, enabling advanced Earth observation tasks including fine-grained land-cover classification [1], [2], [3], [4], quantitative inversion of biophysical parameters [5], [6], [7], [8], [9], and dynamic environmental monitoring. Despite its potential, sensor limitations impose tradeoffs: satellite platforms enable extensive coverage but at coarse spatial resolutions, inadequate for detailed ground object delineation. While a popular solution, hyperspectral super-resolution still requires an initial hyperspectral image [10], [11], [12]; airborne systems deliver high resolution but over limited areas, making them unsuitable for large-scale applications. Consequently, spectral reconstruction from high-resolution multispectral imagery (MSI) has emerged as a viable strategy to enhance spectral resolution while retaining spatial detail and expanding observational utility.

Existing spectral reconstruction methods can be broadly categorized into prior knowledge-based and data-driven approaches. Prior knowledge-based methods establish mappings between MSI and Hyperspectral imagery (HSI) by leveraging mathematical relationships and spectral priors. Representative techniques include dictionary learning and sparse representation [13], [14], [15], [16], [17]. A seminal work in this line is the sparse representation algorithm proposed by Arad and Ben-Shahar [13], which constructs an overcomplete RGB dictionary derived from degraded RGB data and a corresponding RGB training set. It optimizes the sparse representation coefficients of the MSI via sparse coding and directly applies them to a hyperspectral dictionary for reconstruction. However, this approach fails to enforce alignment between the multispectral sparse coefficients and the hyperspectral dictionary, resulting in reduced accuracy. To address this issue, the J-SLOL algorithm jointly optimizes both the sparse coefficients and the dictionary by imposing low-rank constraints and explicitly enforcing coefficient alignment, thereby forming a coupled sparse coding framework for spectral reconstruction

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[15]. Building on this idea, subsequent studies have introduced coupled dictionary learning strategies, in which high- and low-resolution spectral dictionaries are constrained to share sparse coefficients. The alternating direction method of multipliers (ADMMs) is then employed to ensure dictionary alignment [14]. In addition to general priors, the introduction of private spectral libraries has been shown to enhance regional adaptability, while combining complete and private dictionaries further improves reconstruction accuracy [16]. Considering the inconsistency among different spectral libraries, a nonnegative background component–based alignment strategy is adopted to achieve spectral library integration [17]. Beyond dictionary learning, other methods inspired by similar principles have also emerged. One approach reformulates spectral upscaling as a nonlinear mapping problem, leveraging the property that natural spectra lie on a low-dimensional manifold to develop a manifold learning-based technique [18]. Another line of research introduces a Gaussian process prior by clustering spatial–spectral correlated hyperspectral samples, aligning them with RGB images, and applying Beta–Bernoulli process-based Bayesian inference [19]. Nevertheless, these prior knowledge-driven spectral reconstruction methods remain highly dependent on the completeness of prior information, making it difficult to fully recover spectral details in complex land-cover scenarios and limiting their generalization ability. Furthermore, most prior knowledge-based approaches focus primarily on the inherent properties of HSI and spectral libraries, while overlooking the influence of spatial scale effects on the spectral characteristics of ground objects.

In recent years, deep learning has emerged as a transformative paradigm in spectral reconstruction, particularly for HSI from MSI. Pioneering efforts predominantly leveraged convolutional neural networks (CNNs), with residual architectures enhancing feature extraction and surpassing conventional methods in reconstruction fidelity [20]. Subsequent advancements incorporated pyramid-based frameworks to capture multiscale contextual information for mapping sparse spectral bands to comprehensive HSI data [21], alongside parallel CNN modules that bolster joint spatial–spectral representations [22]. However, the intrinsic limitations of CNNs—stemming from fixed kernel sizes and localized receptive fields—hinder their capacity to model long-range dependencies effectively. To address this, self-attention mechanisms have been integrated into spectral reconstruction. For instance, Multihead spectral self-attention modules embedded within U-Net architectures generate attention maps across spectral bands in the spatial domain, thereby accentuating interband correlations and elevating reconstruction quality [23]. The incorporation of feature selection has also become a recent focus of innovation within attention mechanisms [24]. Cross-attention strategies, fusing texture features with spatial–spectral dependencies, have mitigated high-frequency information degradation [25]. Hybrid paradigms combining 3D-CNNs and spectral self-attention further augment local and global modeling, augmented by high-frequency priors and recalibration modules to refine spectral accuracy [26]. Complementing these, many approaches adopt hybrid modeling strategies with an emphasis on physical interpretability. For example, frequency-domain representations derived from discrete cosine transforms, paired with tailored loss functions, regularize learning and enrich spatial–spectral features [27]. Frequency–distance optimization techniques underscore the role of spectral–frequency disparities in robust reconstruction [28]. In addition, Diffusion models grounded in spectral response functions (SRFs) [29] and information-constrained semi-supervised frameworks [30] demonstrate superior efficacy in heterogeneous land-cover environments. Owing to their powerful data distribution modeling and generation capabilities, generative adversarial networks (GANs) have excelled in tasks analogous to reconstruction, such as optical image cloud removal [31] and SAR-optical image fusion and translation [32]. This high level of performance is also evident in their application to spectral reconstruction, such as GANs with advanced feature extractors, which prioritize salient spatial–spectral cues to amplify reconstruction precision [33]. Notwithstanding these strides, persistent challenges impede progress: excessive parameterization in extant models compromises computational efficiency and scalability; underutilization of inherent spatial–spectral priors in HSI data curtails the capture of latent correlations; the frequency domain remains underexplored as a systematic conduit for mitigating spatial–spectral information loss.

To address these challenges, we propose the frequency-enhanced spatial–spectral network (FESSN), a lightweight architecture that synergistically integrates spatial, spectral, and frequency-domain information for efficient MSI-to-HSI reconstruction. Our key contributions are as follows.

1) A novel, parameter-efficient network that fuses multidomain cues through a sequential spatial–spectral–frequency pipeline, achieving superior reconstruction fidelity while maintaining low computational overhead. This design innovates by combining Mamba-attention mechanisms for global-local modeling with frequency modulation, setting a new paradigm for interpretable HSI reconstruction.
2) Mamba-based multiscale spatial fusion module (MMSAF) extracts holistic spatial features by leveraging Mamba to capture long-range dependencies, which guide multikernel depth-wise convolutions for local multiscale integration. It ensures consistent high-frequency detail recovery and contextual awareness, addressing limitations in sparse HSI data.
3) Embedded within a U-shaped network, the U-shaped spectral module (USEM) employs attention mechanisms to capture fine-grained intragroup local correlations and utilizes Mamba to model nonlinear intergroup global interactions. Guided by the sparsity priors of HSI, this structure enables detailed spectral modeling while maintaining physical interpretability.
4) Frequency-enhanced modulation (FEM) that adaptively refines per-pixel spectral amplitudes (for energy and detail control) and phases (for structural preservation) using learnable multilayer perceptrons (MLPs). This introduces global frequency priors with negligible parameter increase, mitigating information loss in blind

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spots and enhancing overall spectral accuracy, as validated by ablations showing amplitude's role in detail recovery and phase's contribution to structural integrity.

The remainder of this article is organized as follows: Section II lays the groundwork by elucidating key technical concepts critical to the study. Section III introduces our innovative FESSN framework, detailing its design and rationale. Section IV presents a comprehensive analysis of extensive experimental results, while Section V delves into insightful discussions. The article concludes with key takeaways in Section VI.

II. Preliminaries

To ensure a comprehensive understanding of the proposed model's foundational elements, the key techniques and concepts underlying the proposed model are introduced.

A. Fourier Frequency-Domain Information

Fast Fourier transform (FFT) and its inverse transform principles provide the theoretical foundation and conceptual guidance for subsequently transferring discrete spectra into the frequency domain for fine-tuning. The 1-D FFT enables a simple and efficient conversion from the spectral domain to the frequency domain. Its mathematical definition is as follows:

$$X_k = \sum_{n=0}^{N-1} x_n \cdot e^{-j2\pi \frac{nk}{N}} \tag{1}$$

where x_n denotes the nth sample point in the spectral domain, X_K represents the corresponding kth frequency component in the frequency domain, and N is the sequence length. This transform essentially projects the original signal onto a set of complex exponential basis functions, revealing its amplitude and phase components at different frequencies.

Since directly computing the discrete Fourier transform (DFT) requires $O(N^2)$ complex multiply-add operations and is therefore inefficient, this work employs the FFT algorithm to reduce the computational complexity to $O(N \log N)$. The FFT is based on the Cooley-Tukey divide-and-conquer strategy, which decomposes a long-sequence transform into several shorter sequence transforms and recursively combines these subtransforms to improve efficiency [34].

In practical implementation, the input 1-D real-valued sequence $x \in \mathbb{R}^N$ is first mapped to the complex domain, and then a 1-D FFT is performed to obtain the complex spectrum $X \in \mathbb{R}^N$. The magnitude of it represents the frequency amplitude, while the phase angle represents the phase information. After feature extraction or modulation in the frequency domain, the signal is typically transformed back to the spectral domain using the 1-D inverse FFT (IFFT)

$$x_n = \frac{1}{N} \sum_{k=0}^{N-1} X_k \cdot e^{j2\pi \frac{nk}{N}}. \tag{2}$$

This parameter-free frequency-domain paradigm inherently incorporates a global receptive field and frequency-structure priors, proving invaluable for analyzing high-dimensional signals like hyperspectral curves by capturing both local details and long-range dependencies.

B. Mamba

Mamba [40] is a recent state-space model (SSM) that builds upon the classical system-theoretic foundation of continuous-time linear dynamical systems. Traditional SSMs, originally studied in cybernetics, describe the evolution of a hidden state $h(t) \in \mathbb{R}^d$ under the influence of an input sequence $x(t)$, governed by the following differential equations:

$$h'(t) = \Delta h(t) + Bx(t) \tag{3}$$

$$y(t) = Ch'(t) + Dx(t) \tag{4}$$

where $\Delta \in \mathbb{R}^{d\times d}$ is the state transition matrix, $B \in \mathbb{R}^{d\times 1}$ is the input matrix that controls how the input influences state changes, $C \in \mathbb{R}^{1\times d}$ is the output matrix that governs how the current state affects the output, and $D \in \mathbb{R}^{1\times d}$ directly expresses how the input influences the output. To adapt this continuous-time formulation to deep learning frameworks, the system is discretized using zero-order hold (ZOH) methods [23], resulting in the widely used structured state-space for sequence modeling (S4) model

$$h_k = \bar{\Delta} h_{k-1} + \bar{B} x_k \tag{5}$$

$$y_k = Ch_k + Dx_k. \tag{6}$$

Here, h_k is the input feature, k indexes the sequence position, and the discretized matrices $\bar{\Delta}$ and $\bar{B}$ are typically approximated using matrix exponentiation and series expansion techniques. This formulation allows the model to capture long-range dependencies across the input sequence with linear time complexity through structured state transitions. After discretization, the model enables a more efficient inference process compared to the standard Transformer architecture.

The Mamba model builds upon the efficient recursive structure of S4 and introduces two key innovations: the input-driven selective mechanism (selective state-space mechanism) and hardware-aware efficient design. These enhancements significantly improve the model's expressive power and computational efficiency in long-sequence modeling.

One of Mamba's core innovations lies in making the state-space modeling parameters dynamic rather than static, driven by the current input content. In particular, for each position x_k in the image sequence, the state update process is defined as

$$h_{k+1} = \Delta(x_k) h_k + B(x_k) x_k, y_k = C(x_k) h_k \tag{7}$$

$$\mathbf{B}(x) = \text{Linear}_{\mathbf{B}}(x)$$

$$\mathbf{C}(x) = \text{Linear}_{\mathbf{C}}(x)$$

$$\Delta(x) = \tau_\Delta(\text{Linear}_1(x) + s_\Delta). \tag{8}$$

In Mamba, the state transition matrix Δ, input mapping B, and output mapping C are all dynamically generated from the current input feature x through three independent linear layers. The softmax function τ_Δ is applied to ensure numerical stability, and s_Δ is a learnable bias term. All parameters are generated sequentially according to the position in the sequence, allowing the model to adapt dynamically to sequence variations. Due to this selective mechanism, Mamba achieves superior performance on a variety of long-sequence modeling tasks while maintaining high efficiency. It has demonstrated strong results not only on language tasks

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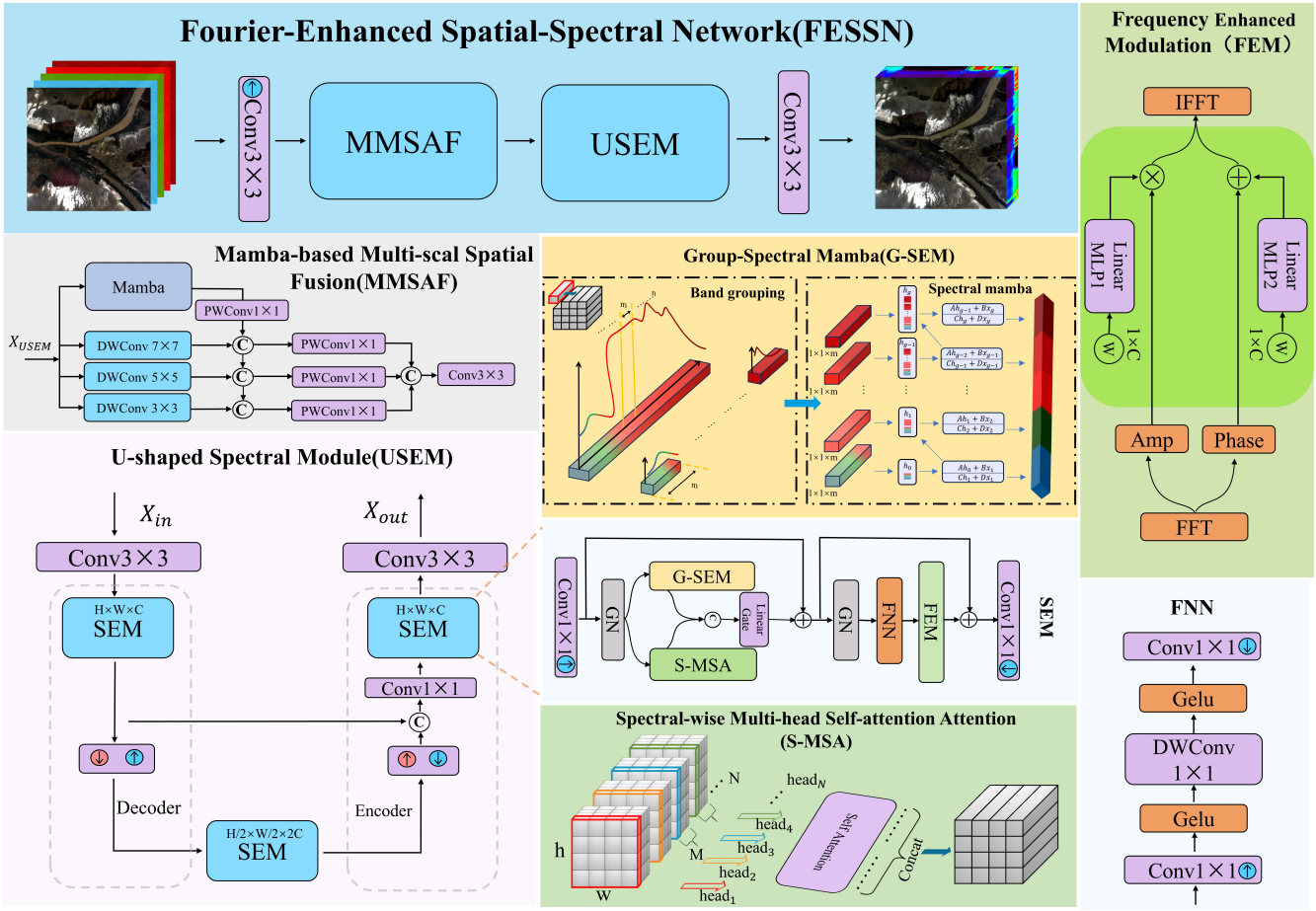

Fig. 1. Overall structure of FESSN for spectral reconstruction. The embedded SEM represents the spectral module, while the FNN denotes the feed-forward neural network. The blue arrows indicate spectral resampling, and the red arrows represent spatial resampling.

but also on computer vision applications, such as semantic segmentation [35] and image classification [2], [36], [37].

III. Proposed Methods

A. Network Architecture

The overall framework of the proposed FESSN model, including detailed structures of all modules, is shown in Fig. 1. To fully learn the spectral complexity of HSI, the input MSI is first up-projected to HSI through a 3×3 convolution, which keeps the spatial resolution unchanged while increasing the spectral dimension to match that of the HSI. All subsequent network modules operate on the hyperspectral dimensionality or even higher. The proposed FESSN model consists of two main parts, following a spatial-then-spectral processing strategy. The first part, MMSAF, is responsible for extracting multiscale spatial features and performing multilevel channel fusion. The second part, the spectral enhancement module (SEM), adopts a U-shaped architecture combined with intra-group and intergroup spectral modeling to capture multiscale spectral features, and integrates a frequency-domain modulation module to avoid the local optimum problem that may occur when modeling only in the spectral domain. The overall workflow of the model is illustrated as follows:

$$F_{\mathrm{HSI}} = \mathrm{SEM}\left(\mathrm{MMSAF}\left(F_{\mathrm{MSI}}\right)\right). \tag{9}$$

By sequentially integrating MMSAF and SEM, the network is capable of comprehensively learning spatial-, spectral-, and frequency-domain information, thereby achieving multidomain joint modeling. This design alleviates the generation of pseudo curves caused by poor reconstruction fitting when relying on a single feature extraction strategy. Finally, a 3×3 convolution is employed to restore both the spatial resolution and spectral dimensionality of the output, enabling the network to flexibly produce HSI with an arbitrary number of spectral bands.

B. Mamba-Based Multiscale Spatial Fusion

To enhance the multiscale modeling capability of spatial features, we design MMSAF

$$F_0 = \mathrm{PWConv}_{1\times1}\left(\mathrm{Mamba}\left(X_{\mathrm{In}}\right)\right) \tag{10}$$

$$F_k = \mathrm{PWConv}_{1\times1}\left(\mathrm{Concat}\left(\mathrm{DWConv}_{k\times k}\left(X_{\mathrm{in}}\right), F_0\right)\right) \tag{11}$$

$$F_{\mathrm{out}} = \mathrm{Conv}_{3\times3}\left(\mathrm{Concat}\left(F_0, F_3, F_5, F_7\right)\right). \tag{12}$$

Here, F_0 represents the processed global contextual feature, and F_k denotes the fused feature obtained by combining the global feature F_0 with the convolutional feature of kernel size k(where $k \in \{3, 5, 7\}$). This module consists of a multibranch structure composed of Mamba and convolutions with different kernel sizes. By performing multilevel parallel fusion of global features and multireceptive-field features,

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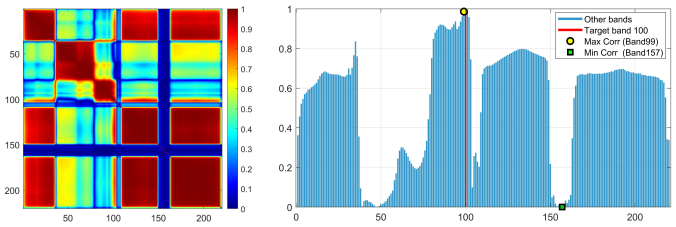

(a) (b)

Fig. 2. Strong local correlations and moderate long-range dependencies among spectral bands. (a) Linear correlation matrix among all spectral bands in the IP dataset. (b) Correlation values between the target band (Band 100) and all other bands.

it forms a global-information-guided multiscale receptive-field enhancement network. Finally, a 3×3 convolution is applied to integrate all features and produce the final output.

C. U-Shaped Grouping Spectral Block

To achieve efficient spectral reconstruction modeling, we propose the USEM, whose core lies in introducing a dual-branch SEM to jointly model both intragroup and intergroup spectral dependencies, with a gating mechanism for adaptive fusion control. The overall framework adopts a single-stage U-Net structure consisting of a symmetric encoder, decoder, and bottleneck, enabling spectral feature extraction and fusion at multiple scales. As shown in Fig. 2, since most remote sensing images exhibit more long-range linear dependencies between bands while local bands tend to be highly correlated, the local and long-range spectral dependencies are modeled separately, and a gating-weighted mechanism is employed for adaptive selection between near and distant dependencies.

The SEM is designed to jointly capture fine-grained relationships along the spectral dimension and the capability for cross-group modeling. To this end, the input features $X_{\text{Input}} \in \mathbb{R}^{C\times H\times W}$ are first divided into N groups, referred to as tokens. To meet the dimensionality requirements for group modeling, the features are expanded to a dimension of $C' = N \cdot d$ through an up-projection operation (Pad), where each group contains $d_h = C'/N$ channels, forming the padded representation

$$X_{\text{pad}} = \text{Pad}\left(X_{\text{Input}}\right), X_{\text{pad}} \in \mathbb{R}^{C'\times H\times W}. \tag{13}$$

The features are then fed into two parallel branches.

1) Local Modeling Branch: Since adjacent hyperspectral bands exhibit strong linear correlations, we adopt a multispectral multihead self-attention mechanism (MS-MSA) for local spectral modeling. In this branch, attention is applied within each head—equivalently, within each spectral group—to model the local interband dependencies in the spectral domain. Given the input spectral features $X_{\text{In}} \in \mathbb{R}^{H\times W\times C'}$, they are first reshaped into $X \in \mathbb{R}^{C'\times HW}$, Subsequently, X is linearly projected into the query $Q \in \mathbb{R}^{C'\times HW}$, key $K \in \mathbb{R}^{C'\times HW}$, and value $V \in \mathbb{R}^{C'\times HW}$ matrices through learnable parameters $W^Q, W^K, W^V \in \mathbb{R}^{C'\times C'}$, enabling the computation of attention weights for local spectral dependency modeling

$$Q = XW^Q, \quad K = XW^K, \; V = XW^V. \tag{14}$$

Since hyperspectral spectra exhibit pronounced fine-grained variations within certain bands, a multihead approach is employed to model the intragroup local spectral dependencies. In this process, the features are split along the spectral dimension into N heads, resulting in $Q = [Q_1, \ldots, Q_N], K = [K_1, \ldots, K_N], V = [V_1, \ldots, V_N]$, where each head has a dimensionality of $d_h = C'/N$. Each head is then processed through a self-attention mechanism to compute attention weights, followed by the concatenation of all weighted heads to form the updated representation, as expressed by

$$A_j = \text{softmax}\left(\sigma_j K_j^T Q_j\right), \quad \text{head}_j = V_j A_j \tag{15}$$

$$X_{\text{Out}} = \left(\text{Concat}_{j=1}^{N}\left(\text{head}_j\right) W\right) + \text{PE}\left(V\right). \tag{16}$$

Here, $\sigma_j \in \mathbb{R}^1$ is applied to align the interspectral sparsity, and after concatenation, a linear layer $W \in \mathbb{R}^{C\times C}$. Subsequently, positional encoding PE(V) is added to incorporate spectral positional information.

2) Global Modeling Branch: As shown in Fig. 2, besides the strong linear correlations among local spectral bands, certain distant spectral bands also exhibit linear dependencies. In addition, nonlinear interband correlations remain to be further explored. To address this, we design the Spectral Mamba module, which leverages the SSM to perform sequential modeling across spectral groups. In particular, the input features along the spectral dimension are divided into N groups, and the standard Mamba module is applied to model the relationships between different groups. These intergroup relationships are then used to guide the update of spectral features. The overall workflow for intergroup spectral feature updating is as follows:

$$H_G = \text{SplitSpectralGroup}\left(X_{\text{In}}\right) \tag{17}$$

$$H_F = \text{Flatten}\left(H_G\right) \tag{18}$$

$$H_M = \text{Mamba}\left(H_F\right) \tag{19}$$

$$H_{\text{Output}} = \text{Reshape}\left(H_M\right) + X_{\text{In}}. \tag{20}$$

Here, $H_G \in \mathbb{R}^{H\times W\times N\times M}, H_F \in \mathbb{R}^{HW\times N\times M}, H_M \in \mathbb{R}^{HW\times N\times M}$, and $H_{\text{Output}} \in \mathbb{R}^{C'\times H\times W}$ denote the split spectral group features, the stretched features, the features modeled by Mamba, and the residual output features, respectively. N represents the number of spectral groups, M is the number of bands in each spectral group, and Mamba refers to the standard Mamba block.

3) Gating Fusion: The dual-branch outputs are denoted as $X_{\text{S-MSA}} \in \mathbb{R}^{C'\times H\times W}$ and $X_{\text{Mamba}} \in \mathbb{R}^{C'\times H\times W}$, and a two-layer fully connected gating module is employed to predict the fusion weights

$$W = \sigma\left(\text{FC}\left(\text{Concat}\left(X_{\text{S-MSA}}, X_{\text{Mamba}}\right)\right)\right) \tag{21}$$

$$X_{\text{fsd}} = W \odot X_{\text{S-MSA}} + (1 - W) \odot X_{\text{Mamba}} \tag{22}$$

$$X_{\text{result}} = \text{FFN}\left(X_{\text{fsd}}\right) + X_{\text{fsd}}. \tag{23}$$

Here, FC denotes the gating network composed of two linear layers, and σ represents the Sigmoid activation function. This mechanism adaptively determines the relative contribution of local versus global modeling, assigning different preferences to different regions.

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D. Frequency-Enhanced Modulation

To enhance the spectral fidelity of reconstructed HSI, we propose an FEM, which refines the output spectral features in the frequency domain. This module enables independent adjustment of both the amplitude and phase components of the spectral signals, effectively correcting frequency distortions introduced during network reconstruction and enhancing the continuity and smoothness of the spectra.

Let the reconstructed HSI output by the network be $X \in \mathbb{R}^{C\times H\times W}$, a 1-D FFT is applied along the spectral dimension C to obtain its complex-valued frequency-domain representation

$$F = Fc(X) = A \cdot e^{j\Phi}. \tag{24}$$

Here, $A = |F| \in \mathbb{R}^{C\times H\times W}$ denotes the spectral amplitude and $\Phi = \angle F \in \mathbb{R}^{H\times W\times C}$ denotes the phase spectrum.

Subsequently, lightweight MLPs are employed to generate the amplitude modulation parameters and phase-shifted parameters, respectively,

$$S_A = \mathrm{MLP}_A(1) \in \mathbb{R}^{1\times C} \tag{25}$$

$$S_\Phi = \mathrm{MLP}_\Phi(0) \in \mathbb{R}^{1\times C}. \tag{26}$$

Each MLP consists of several linear layers with ReLU activations, and its output dimensionality matches the number of spectral bands C. To ensure stable training in the early stages, we adopt the following initialization strategy for the modulation parameters: the amplitude modulation vector S_A is initialized to 1 to preserve the original amplitude structure, whereas the phase modulation vector S_Φ is initialized to 0 to avoid introducing phase perturbations at initialization. After broadcasting these parameters in the frequency domain and applying the modulation, we obtain

$$A' = A \cdot S_A \tag{27}$$

$$\Phi' = \Phi + S_\Phi \tag{28}$$

$$F' = A' \cdot e^{j\Phi'} = A'\left(\cos\Phi' + j\sin\Phi'\right). \tag{29}$$

Finally, the IFFT is applied to restore the data to the original spatial–spectral domain:

$$X' = \mathcal{F}_c^{-1}\left(F'\right). \tag{30}$$

This frequency-domain modulation strategy can be regarded as a learnable frequency correction mechanism that finely rectifies spectral distortions. Since the modulation parameters are global vectors, the overall number of parameters is extremely low, introducing negligible additional computational cost while effectively improving spectral reconstruction accuracy.

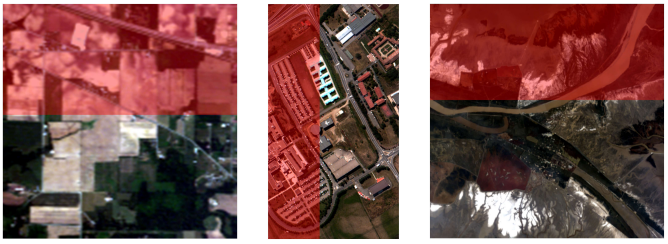
IV. Experiments

A. Datasets

The Indian Pines (IP) dataset was acquired by the AVIRIS hyperspectral sensor, with a spatial resolution of 20 m, covering 220 spectral bands in the wavelength range of 400–2500 nm. The corresponding ideal MSIs were simulated via the Sentinel-2 SRF, containing 13 bands. The entire scene has a size of 145 × 145 × 220, where a subregion of 145 × 65 × 220 was selected for training. A sliding window approach with a stride of 4 was employed to generate 105 patches of 64 × 64 as training samples, and testing was performed on the full image. The dataset contains a total of 10 249 labeled samples spanning 16 annotated land-cover classes, such as Alfalfa, Corn.

TABLE I
Details of Datasets Used in This Article

Dataset sign	Platform	Image size	Spatial resolution	Aquisition date
IP	Airplain	145×145×220	20m	1992
PU	Airplain	610×340×103	1.3m	2002
GF-5	Satellite	500×500×140	30m	Novermber 1, 2018



(a) (b) (c)

Fig. 3. Datasets introduction. (a) IP. (b) PU. (c) GF-5. The red mask indicates the training region.

The Pavia University (PU) dataset was collected by the ROSIS hyperspectral sensor, with a spatial resolution of 1.3 m, covering 103 spectral bands within the 430–860-nm range. The ideal MS data were simulated using the Sentinel-2 SRF, selecting the first nine bands based on the PU spectral range. The full dataset has a size of 610 × 340 × 103, and a subregion of 250 × 340 × 103 was used for training. With a sliding window of stride 16, 216 patches of 64 × 64 were generated as training samples, while evaluation was conducted on the entire scene. The dataset contains 7320 labeled samples, covering nine land-cover categories such as Asphalt, Meadows, and Gravel.

The GaoFen-5 (GF-5) dataset was obtained from the hyperspectral sensor onboard the GF-5 satellite, with a spatial resolution of 30 m. After removing invalid bands affected by water vapor absorption, low signal-to-noise ratio, and abrupt reflectance changes in the atmosphere, 140 valid spectral bands in the 390–983-nm range were retained. The ideal MS data were simulated using the Sentinel-2 SRF, selecting the first ten bands according to the GF-5 spectral range. The dataset size is 500 × 500 × 140, with a subregion of 500 × 200 × 140 used for training. A sliding window with a stride of 16 generated 252 patches of 64 × 64 for training, and testing was conducted on the full scene. In total, the dataset contains 250 000 labeled samples covering eight land-cover categories, including Spartina alterniflora, Reed, and Tamarix chinensis forest, among others. The specific parameters of the datasets and the corresponding training ratios are summarized in Table I and Fig. 3, respectively.

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B. Environment Configuration and Experimental Settings

All models were implemented and trained using the PyTorch framework. Training was performed on an NVIDIA RTX 4090 GPU for 200 epochs, during which the evaluation metrics gradually converged and stabilized. The overall training across the four datasets required approximately 32 min in total. For the experimental settings, the batch size was set to 10, and the Adam optimizer was employed with an initial learning rate of 0.0002. The learning rate was decayed following a cosine annealing schedule. During training, overlapping HSI–MSI sample pairs of size 64×64 were randomly cropped from the original images and augmented via fixed-stride sliding window extraction, random flipping, and other data augmentation techniques before being fed into the network. During testing, the performance of different models was evaluated on all 64×64 patches from the entire image. In particular, the testing area was cropped into overlapping MS patches of size 64×64 with a stride of 64, each reconstructed into the corresponding HS patch, and then the final HS image was obtained by stitching all reconstructed patches together. In both the G-SEM and S-MSA modules, the number of spectral groups was set to 8. In the FEM, both amplitude and phase modulation were enabled

$$\mathcal{L}_{\text{RMSE}}\left(Y, \hat{Y}\right) = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left(Y - \hat{Y}\right)^2}. \tag{31}$$

For similarity-based evaluation, we assessed the performance of different methods from both quantitative and qualitative perspectives. Quantitative metrics include the root-mean-squared error (RMSE) [38], peak signal-to-noise ratio (PSNR) [39], spectral angle mapper (SAM) [40], relative global dimensional error in synthesis (ERGAS) [41], and structural similarity index measure (SSIM) [42].

Experimental results show that, compared with the $\ell 2$-norm MSE, using RMSE as the loss function to constrain the entire modeling process yields better similarity evaluation performance. This may be attributed to the fact that RMSE is more robust to extreme errors and shares the same unit as the original spectra. Moreover, RMSE is less sensitive to outliers caused by noise in individual hyperspectral bands, which may facilitate the capture of fine details during optimization.

From a qualitative perspective, performance differences among methods were visualized using error maps. In addition, spectral curves of nine different land-cover classes were sampled from the three datasets for comparative analysis. For the classification experiments, three evaluation metrics widely used in HSI classification were adopted, including overall accuracy (OA), average accuracy (AA), and the Kappa coefficient (κ).

C. Similarity Evaluation

In the testing phase, we conducted a quantitative comparison of the proposed FESSN against six state-of-the-art models, including HDNET [28], AWAN [43], HSCNN [20], MST++ [23], MSFN [44], and RepCPSI [45]. The quantitative evaluation results on the three datasets are presented in Tables II–IV, where the best results are highlighted in bold, and the second-best results are underlined. As shown, our proposed FESSN consistently outperforms all competing methods across all evaluation metrics. In particular, compared with the second-best results, FESSN achieves relative improvements of 5.30%, 4.95%, and 5.31% on RMSE, SAM, and ERGAS, respectively, for the IP dataset; 2.86%, 2.45%, and 2.91% for the PU dataset; and 2.36%, 2.86%, and 2.73% for the GF-5 dataset. In addition, noticeable improvements are also observed on the remaining metrics, further demonstrating the robustness of FESSN as well as its adaptability across diverse scenarios.

To provide a more intuitive evaluation of reconstruction performance across different models, we present both error maps and single-pixel spectral curve visualizations on the IP, PU, and GF-5 datasets. In the error maps, colder color tones correspond to lower RMSE values, indicating more

TABLE II
COMPARISON OF SIMILARITY METRICS ON THE ID DATASET

Method	RMSE(↓)	PSNR(↑)	SAM(↓)	ERGAS(↓)	SSIM(↑)
HDNET	0.003448	48.782646	0.011009	1.349821	0.993370
AWAN	0.003701	48.162655	0.011462	1.449278	0.993081
HSCNN	0.003240	49.393318	0.010316	1.268988	0.994471
MST++	0.003211	49.419048	0.010246	1.256772	0.994238
MSFN	0.003439	48.856159	0.010886	1.346286	0.993634
RepCPSI	0.003057	49.819859	0.009829	1.196548	0.988877
Ours	**0.002897**	**50.297153**	**0.009347**	**1.133721**	**0.995159**

The best results are marked in **bold**, and the second-best results are underlined.

TABLE III
COMPARISON OF SIMILARITY METRICS ON THE PU DATASET

Method	RMSE(↓)	PSNR(↑)	SAM(↓)	ERGAS(↓)	SSIM(↑)
HDNET	0.004653	45.755932	0.031572	3.489264	0.991052
AWAN	0.005098	44.714394	0.034150	3.822308	0.989877
HSCNN	0.004766	45.137745	0.032291	3.575063	0.990303
MST++	0.004700	45.388668	0.031854	3.523402	0.990699
MSFN	0.004748	45.417953	0.032056	3.560184	0.990744
RepCPSI	0.004614	45.858227	0.031292	3.460868	0.991404
Ours	**0.004477**	**46.268200**	**0.030515**	**3.356762**	**0.991533**

The best results are marked in **bold**, and the second-best results are underlined

TABLE IV
COMPARISON OF SIMILARITY METRICS ON THE GF-5 DATASET

Method	RMSE(↓)	PSNR(↑)	SAM(↓)	ERGAS(↓)	SSIM(↑)
HDNET	0.004947	45.528118	0.020113	2.341998	0.993876
AWAN	0.005290	45.055996	0.021215	2.503203	0.993540
HSCNN	0.005212	45.068485	0.021180	2.311440	0.993790
MST++	0.004890	45.752052	0.019709	2.311440	0.994081
MSFN	0.004916	45.708172	0.019751	2.328213	0.994057
RepCPSI	0.004832	45.686680	0.019875	2.288154	0.994109
Ours	**0.004716**	**45.994789**	**0.019211**	**2.224558**	**0.994292**

The best results are marked in **bold**, and the second-best results are underlined.

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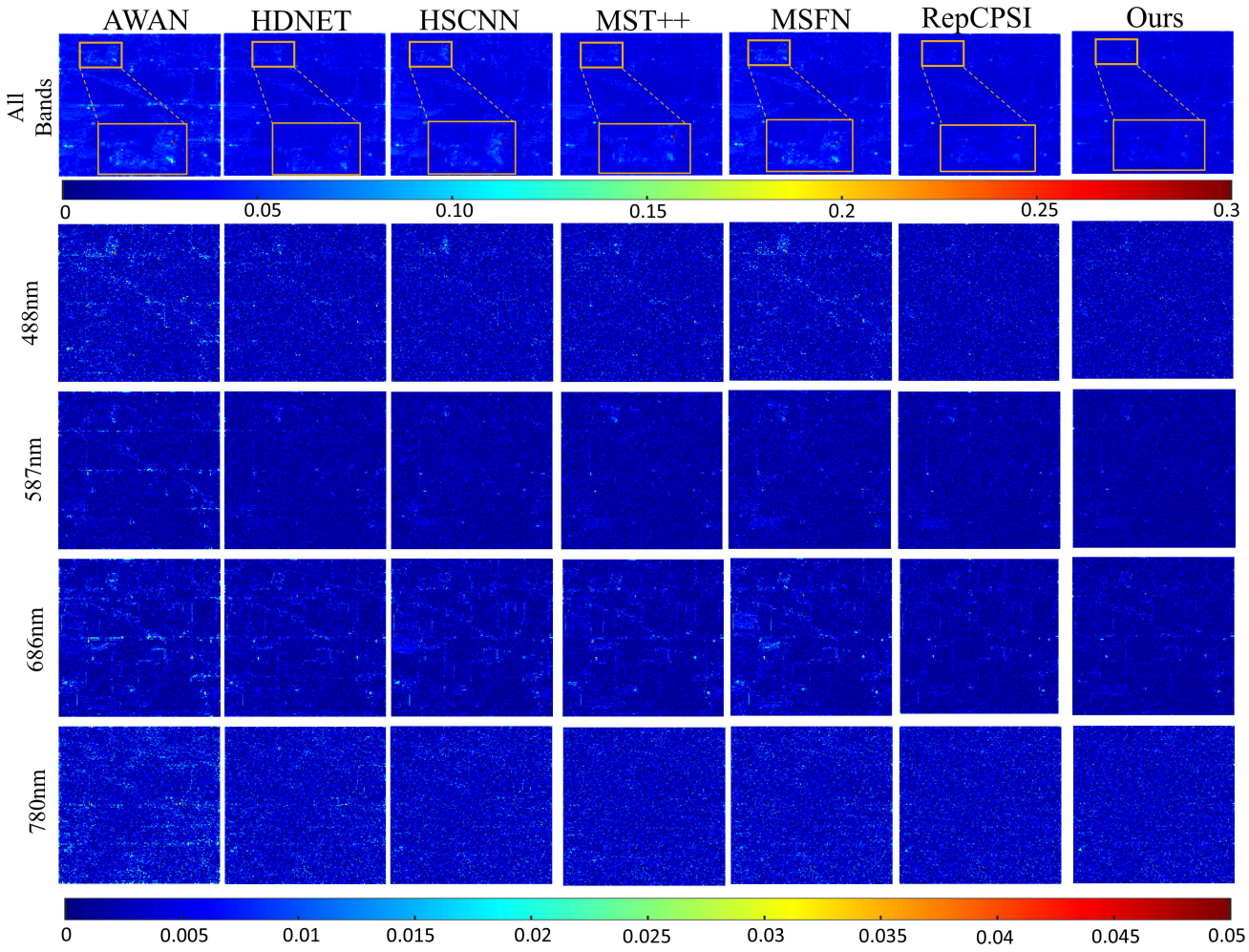

Fig. 4. Error map on the IP dataset.

faithful reconstructions. We also compare the reconstruction performance not only across all spectral bands but also on several key individual bands that are commonly used in remote sensing analysis, including red, green, blue, and near-infrared. Across all three datasets, FESSN consistently achieves superior reconstruction quality compared with the six competing models. To better illustrate fine-grained reconstruction details, we highlight enlarged regions where most baseline models exhibit inferior performance.

On the IP dataset, RepCPSI achieves relatively good reconstruction performance. It introduces a novel CNN-based architecture with reparameterization techniques to model local spectral dependencies. In particular, the local spectral modeling assumes that hyperspectral bands exhibit high correlations only in local ranges, while the correlations with distant bands are negligible and thus considered redundant. Based on this assumption, RepCPSI employs convolution operations along the vertical and horizontal spatial dimensions for local spectral modeling. However, in geoscience studies, such as the IP dataset, where the spectral range extends beyond 2000 nm, the dependencies among spectral bands are far more complex. Long-range dependencies become indispensable for accurate modeling, which makes RepCPSI less suitable for geoscientific applications. As shown in Table II and Fig. 4, our experimental results demonstrate that the proposed FESSN achieves the best performance in both quantitative and qualitative evaluations on the IP dataset.

On the PU dataset, HDNET also recognizes the importance of frequency-domain signal modulation and achieves promising reconstruction results. It exploits frequency-domain differences based on a frequency–distance optimization method. However, this process is only applied in the loss function design and does not explicitly modulate frequency-domain information. In contrast, the proposed FEM module directly operates in the frequency domain of each pixel, initialized with no modulation, and dynamically updates the amplitude and phase parameters through backpropagation. This strategy explicitly optimizes the spectral-frequency characteristics while ensuring a more interpretable and intuitive parameter update process. By enlarging the high-frequency regions in the error maps for detailed analysis, as shown in Fig. 5 and Table III, our model demonstrates superior performance in restoring high-frequency edge information while simultaneously maintaining state-of-the-art reconstruction quality in background low-frequency regions.

On the GF dataset, compared with MST++, which demonstrates competitive reconstruction performance, our method also employs a U-shaped structure for spectral feature extrac-

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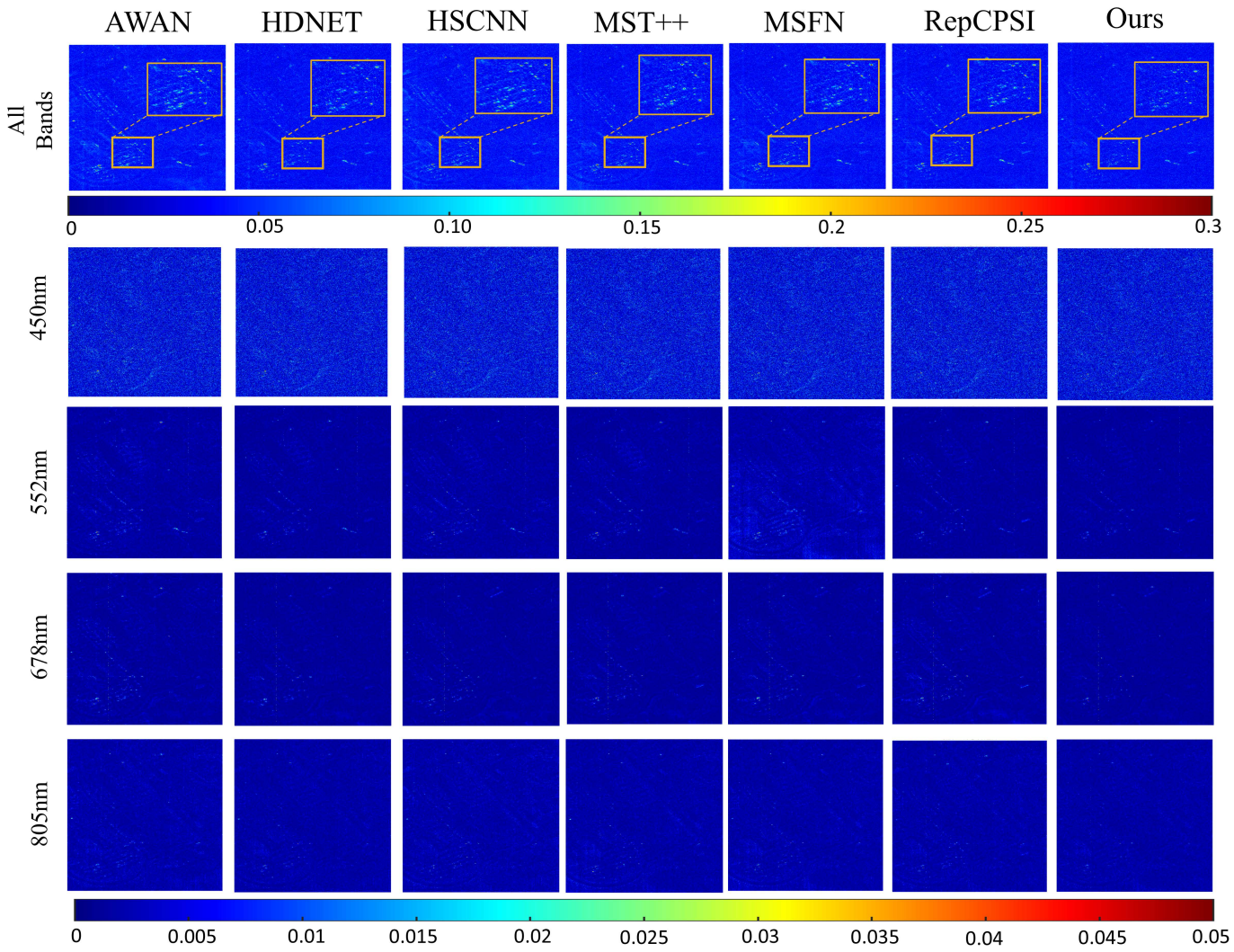

Fig. 5. Error map on the PU dataset.

TABLE V
ABLATION STUDY ON THE EFFECTIVENESS OF EACH MODULE IN FESSN ON THE IP, PU, AND GF-5 DATASETS

Modules Ablation			IP dataset			PU dataset			GF-5 dataset		
MMSAF	USEM	FEM	RMSE(↓)	PSNR(↑)	SAM(↓)	RMSE(↓)	PSNR(↑)	SAM(↓)	RMSE(↓)	PSNR(↑)	SAM(↓)
√			0.003134	49.606892	0.009905	0.004749	45.527603	0.031805	0.005313	44.899715	0.022020
	√		0.003163	49.546352	0.010134	0.004642	45.851849	0.031448	0.004872	45.722900	0.019847
√	√		0.003019	49.930840	0.009690	0.004538	45.973728	0.030941	0.004787	45.855129	0.019478
√	√	√	**0.002895**	**50.309631**	**0.009342**	**0.004482**	**46.285358**	**0.030525**	**0.004718**	**46.012928**	**0.019145**

The best results are marked in **bold.**

tion. However, by incorporating a multiscale spatial feature extraction module before spectral feature learning, together with the embedded spectral–frequency modulation module, our approach achieves a significant improvement in reconstruction accuracy. Moreover, we adopt only a two-layer U-shaped structure combined with lightweight modules specifically tailored for HSI reconstruction. As reported in Table IV and the table of model complexity in the discussion section, our model contains merely 37.52% of the parameters of MST++, yet achieves an RMSE improvement of 3.56%. As shown in Fig. 6, we highlight high-reflectance regions, which are generally less stable and pose greater challenges for reconstruction. It can be observed that our model consistently yields superior performance in these regions. In addition, error analysis conducted on four representative bands further confirms that our method achieves the best reconstruction

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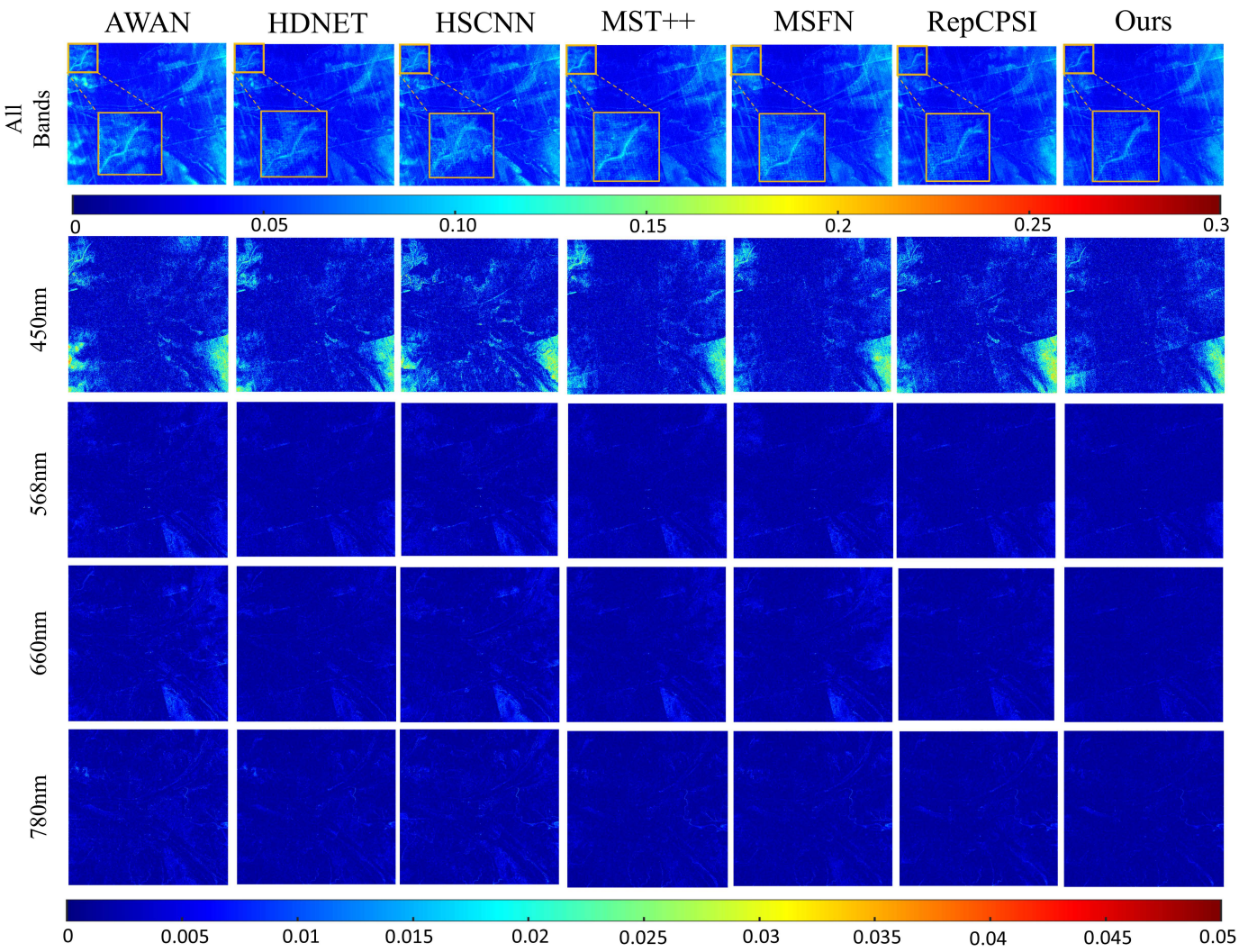

Fig. 6. Error map on the GF-5 dataset.

TABLE VI

OVERALL PERFORMANCE UNDER DIFFERENT MODULE REPLACEMENT STRATEGIES ON THE IP, PU, AND GF-5 DATASETS

Dataset	Modules	RMSE(↓)	PSNR(↑)	SAM(↓)
ID	S^2F-FAB	0.003040	49.880222	0.009739
	FP	0.003083	49.741516	0.009873
	Ours	**0.002895**	**50.309631**	**0.009342**
PU	S^2F-FAB	0.004665	45.881264	0.031539
	FP	0.004565	45.880981	0.031103
	Ours	**0.004482**	**46.285358**	**0.030525**
GF	S^2F-FAB	0.005205	45.102798	0.021574
	FP	0.004810	45.823246	0.019608
	Ours	**0.004718**	**46.012928**	**0.019145**

The best results are marked in **bold.**

TABLE VII

EFFECTIVENESS OF MMSAF MODULATION ON THE IP, PU, AND GF-5 DATASETS

Dataset	Modules	RMSE(↓)	PSNR(↑)	SAM(↓)
IP	Normal	0.003065	49.826107	0.009852
	MMSAF(Local)	0.002939	50.167091	0.009479
	MMSAF(Global)	**0.002895**	**50.309631**	**0.009342**
PU	Normal	0.004541	45.997791	0.030919
	MMSAF(Local)	0.004540	45.870205	0.030836
	MMSAF(Global)	**0.004482**	**46.285358**	**0.030525**
GF-5	Normal	0.004832	45.814526	0.019650
	MMSAF(Local)	0.004836	45.799984	0.019611
	MMSAF(Global)	**0.004718**	**46.012928**	**0.019145**

The best results are marked in **bold.**

results. For the single-pixel spectral curve visualizations, three land-cover classes are sampled from each dataset, resulting in a total of nine categories. As shown in Fig. 7, we compare the performance of different models across various land-cover types by examining both the overall and locally magnified spectral curves and evaluating them using the SAM metric. Again, FESSN demonstrates consistently better reconstruction performance than the six competing models on all three datasets.

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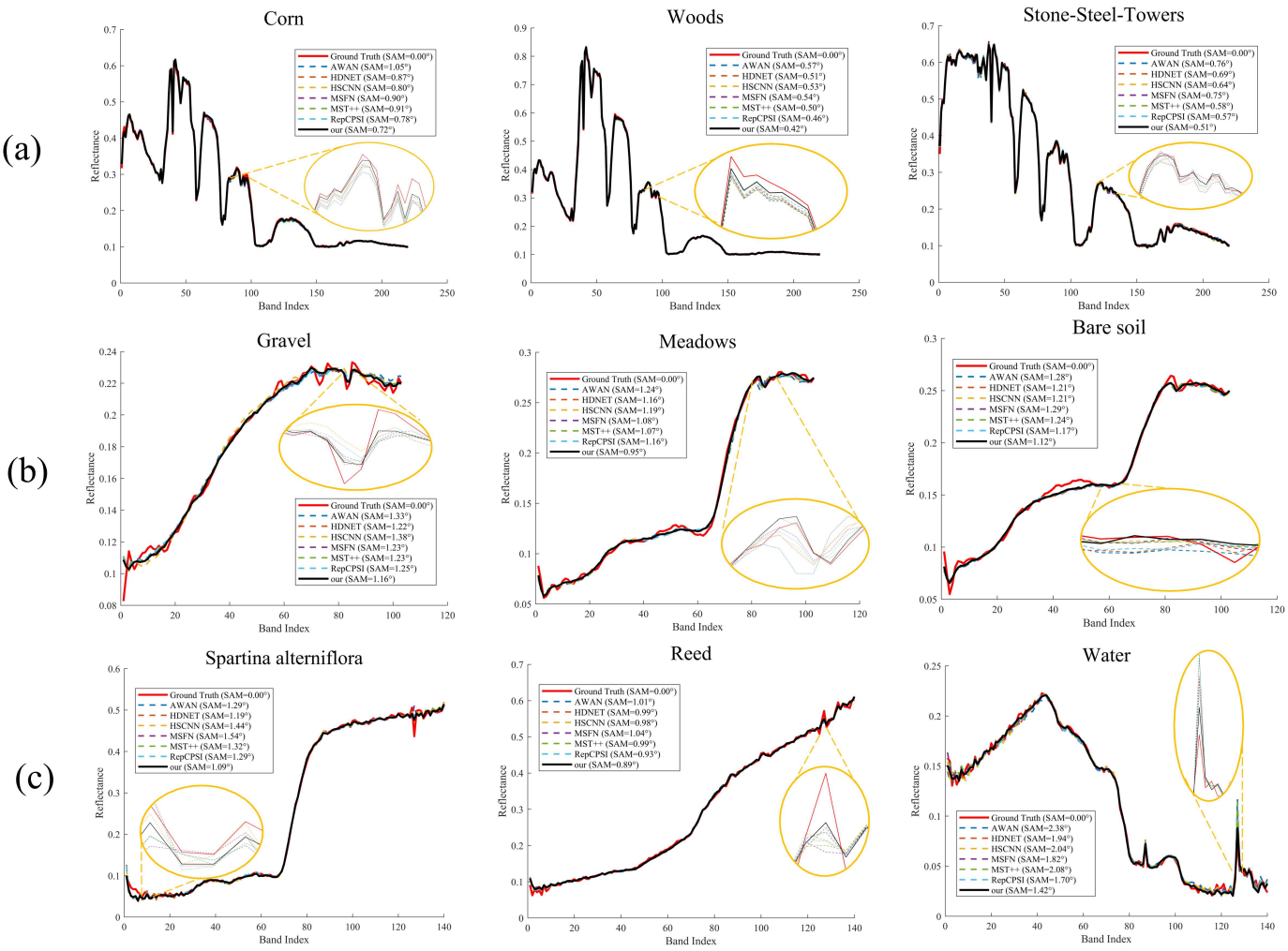

Fig. 7. Spectral curve and SAM comparison on nine classes across three datasets. (a) IP dataset. (b) PU dataset. (c) GF-5 dataset.

TABLE VIII

EFFECTIVENESS OF THE SPECTRAL MAMBA MODULE ON THE IP, PU, AND GF-5 DATASETS

Dataset	Modules	RMSE(↓)	PSNR(↑)	SAM(↓)
IP	Attention	0.002938	50.178738	0.009466
	Attention+Mamba	**0.002895**	**50.309631**	**0.009342**
PU	Attention	0.004503	46.042953	0.030633
	Attention+Mamba	**0.004482**	**46.285358**	**0.030525**
GF-5	Attention	0.004752	45.949028	0.019325
	Attention+Mamba	**0.004718**	**46.012928**	**0.019145**

The best results are marked in **bold.**

TABLE IX

EFFECTIVENESS OF THE FREQUENCY-DOMAIN MODULATION ON THE IP, PU, AND GF-5 DATASETS

Dataset	Modules	RMSE(↓)	PSNR(↑)	SAM(↓)
IP	Phase	0.003036	49.887173	0.009741
	Amp	0.002897	50.297153	0.009347
	P+A	**0.002895**	**50.309631**	**0.009342**
PU	Phase	0.004536	46.006622	0.030896
	Amp	**0.004477**	46.268200	**0.030515**
	P+A	0.004482	**46.285358**	0.030525
GF-5	Phase	0.004811	45.826408	0.019560
	Amp	**0.004716**	45.994789	0.019211
	P+A	0.004718	**46.012928**	**0.019145**

The best results are marked in **bold.**

V. DISCUSSION

In this section, we discuss five types of experiments to comprehensively evaluate the model's generalization capability and the effectiveness of each component, including module ablation comparisons, module hyperparameter analysis, Similarity evaluation on real datasets, classification assessment, and computational cost comparison.

A. Module Ablation Experiment

1) Overall Ablation of FESSN: We conducted four groups of experiments on each dataset, as presented in Table V, where both individual modules and their combinations were evaluated to verify their effectiveness. We observed that in relatively simple scenarios, such as the IP dataset, the lightweight MMSAF

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TABLE X
EVALUATION OF SIMILARITY METRICS USING GF-5 AND SENTINEL-2 IMAGES

Method	RMSE(↓)	PSNR(↑)	SAM(↓)	ERGAS(↓)	SSIM(↑)
HDNET	0.035799	29.298817	0.062722	16.659918	0.899863
AWAN	0.038404	29.625010	0.056907	18.473583	0.903828
HSCNN	0.049621	27.439558	0.067209	23.566730	0.874429
MST++	0.033844	29.935797	0.058300	15.727477	0.905347
MSFN	0.036463	30.139490	0.054841	16.556072	0.911227
RepCPSI	0.034688	29.483326	0.059607	16.077303	0.906857
Ours	**0.033677**	**30.440855**	**0.052078**	**15.416440**	**0.916455**

The best results are marked in **bold**, and the second-best results are underlined.

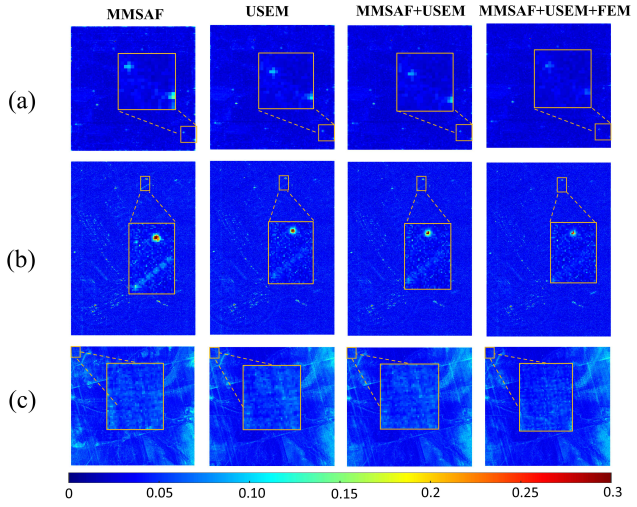

Fig. 8. Ablation study on the effectiveness of each module in FESSN. (a) IP dataset. (b) PU dataset. (c) GF-5 dataset.

spatial information modeling module performs comparably to USEM. However, in more complex scenes like the PU and GF-5 datasets, USEM demonstrates superior performance, indicating that spatial modeling modules alone may struggle in handling more challenging reconstruction scenarios. As shown in Fig. 8, the frequency-domain modulation module effectively smooths large reconstruction errors, thanks to its fine-grained adjustment of high-frequency details, which also enhances the model's robustness to noise. Finally, through error maps and quantitative metrics, we demonstrate that the complete integrated model achieves the best overall reconstruction performance.

2) Comparison With Existing Frequency and Attention Mechanisms: In this section, we conduct replacement experiments to evaluate the effectiveness of our proposed modules within the overall model architecture. Spatio-spectral feature fusion attention blocks (S^2F-FAB) [46] and Fourier domain perception block (FP) [47] are derived from advanced spatial–spectral attention and frequency-perception modules previously proposed in the spatial super-resolution domain. In particular, S^2F-FAB replaces the SEM module in USEM as the core spectral processing component within the U-Net backbone, while FP replaces the FEM module by following its original residual frequency-perception configuration. Table VI

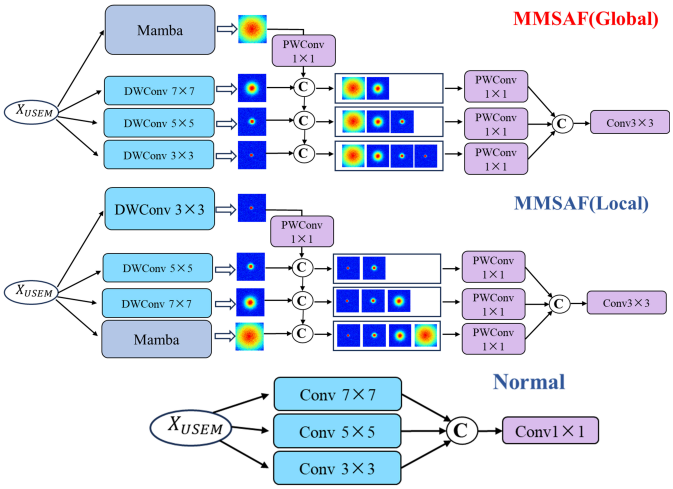

Fig. 9. Illustration of three different spatial feature extraction strategies.

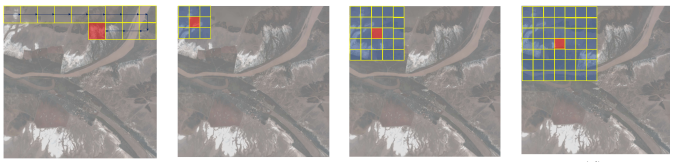

Fig. 10. Comparison between global and local spatial feature extraction using convolutional kernels of different sizes. Four feature extraction processes are illustrated as: (a) Mamba, (b) CNN (k = 3), (c) CNN (k = 5), and (d) CNN (k = 8).

shows that when the SEM spectral processing unit is replaced by the S^2F-FAB module, the RMSE increases by 4.7%, 3.9%, and 9.3% on the IP, PU, and GF datasets, respectively. This indicates that the proposed grouped intraspectral and interspectral attention mechanism achieves more refined and comprehensive spectral modeling than conventional spectral attention. Furthermore, the RMSE increases by 6.0%, 1.8%, and 1.9% on the IP, PU, and GF datasets, respectively, when the FEM is replaced with the FP, further verifying that the proposed frequency modulation mechanism serves as a direct, efficient, and effective spectral refinement component that enables explicit modulation in the frequency domain.

3) Comparison With Existing Multiscale Spatial Fusion: Conventional multiscale convolutional fusion modules integrate features from different convolutional scales and perform 1×1 convolution for channel-wise fusion, but they lack guidance from global information, resulting in feature aggregation limited to local spatial regions. As illustrated by MMSAF(Global) in Fig. 9, we concatenate a Mamba module to the output of each convolutional scale, which models the global dependencies for each pixel. Consequently, this design enables every level of local feature to carry global context, and large-scale information is continuously reused by smaller-scale branches, thereby achieving "global information-guided multiscale spatial–spectral joint modeling." As shown in Fig. 10, the Mamba module allows pixels to capture longer-range dependencies and provides richer spatial semantic information.

To highlight the stability of our architecture, we compared three spatial information aggregation strategies:

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TABLE XI
CLASSIFICATION PERFORMANCE COMPARISON ON THE IP DATASET

Class	Color	HDNET	AWAN	HSCNN	MST++	MSFN	RepCPSI	Ours	R-HSI	F-MSI
1		59.55	59.09	68.18	**72.73**	57.73	68.18	57.27	64.55	21.36
2		76.18	73.01	75.31	74.19	76.49	73.97	74.25	**77.20**	62.71
3		54.20	52.60	50.16	49.73	56.15	53.64	56.30	**63.73**	31.74
4		43.01	52.12	48.32	49.91	41.77	49.1	51.50	**63.54**	26.11
5		90.11	86.36	89.67	86.58	89.76	**90.63**	89.28	89.54	88.76
6		94.70	97.78	97.20	94.93	97.00	95.53	**98.04**	92.88	93.43
7		58.52	66.67	65.93	66.67	78.52	58.52	**85.19**	70.37	7.41
8		93.54	98.11	94.59	97.98	96.44	95.56	96.31	**98.77**	99.21
9		22.11	9.47	31.58	16.84	30.53	28.42	30.53	**40.00**	0.00
10		60.22	63.9	61.26	63.72	60.02	63.14	64.98	**67.81**	47.79
11		81.39	79.9	79.34	81.89	81.7	83.19	83.21	82.19	**83.23**
12		69.47	71.81	73.16	71.88	68.51	71.38	**73.79**	72.06	30.28
13		98.56	98.77	98.36	97.23	96.10	96.31	**98.87**	97.33	97.44
14		97.37	96.59	**97.79**	97.27	97.7	97.32	95.77	95.51	96.99
15		**57.77**	57.17	51.2	50.08	49.32	50.52	53.02	50.74	39.73
16		85.84	87.87	89.66	86.29	79.10	87.64	85.84	84.27	**90.11**
OA(%)		77.99	77.94	77.61	77.91	78.13	78.52	79.27	**80.08**	69.94
AA(%)		71.41	71.96	73.24	72.37	72.30	72.69	74.64	**75.65**	57.27
KAPPA(%)		74.74	74.70	74.32	74.63	74.86	75.31	76.24	**77.20**	64.99

The best results are marked in **bold**, and the second-best results are underlined.

TABLE XII
CLASSIFICATION PERFORMANCE COMPARISON ON THE PU DATASET

Class	Color	HDNET	AWAN	HSCNN	MST++	MSFN	RepCPSI	Ours	R-HSI	F-MSI
1		91.89	92.28	**93.24**	92.19	91.11	91.69	91.40	92.20	93.05
2		97.48	97.59	97.01	97.85	**98.32**	97.85	98.10	97.01	97.44
3		64.79	65.92	66.96	69.95	67.62	70.83	69.37	**71.51**	47.11
4		88.60	90.87	89.14	91.16	89.52	91.62	**91.67**	90.94	83.42
5		99.06	99.20	99.15	**99.27**	99.08	99.20	99.24	99.24	99.21
6		79.63	76.74	72.53	72.32	74.31	74.15	76.18	**80.91**	48.03
7		78.47	80.46	**83.85**	82.50	81.07	70.83	78.93	78.13	0.00±
8		86.71	86.95	**88.15**	85.04	86.15	86.62	87.26	85.34	86.85
9		99.78	99.81	99.85	99.78	99.76	99.87	**99.89**	99.85	99.78
OA(%)		90.86	90.93	90.46	90.63	90.72	90.62	91.16	**91.23**	83.64
AA(%)		87.38	87.76	87.76	87.79	87.44	86.96	88.00	**88.35**	72.77
KAPPA(%)		87.77	87.85	87.21	87.42	87.54	87.42	88.15	**88.29**	77.59

The best results are marked in **bold**, and the second-best results are underlined.

TABLE XIII
CLASSIFICATION PERFORMANCE COMPARISON ON THE GF-5 DATASET

Class	Color	HDNET	AWAN	HSCNN	MST++	MSFN	RepCPSI	Ours	R-HSI	F-MSI
1		83.94	86.95	84.01	86.48	**88.58**	87.29	88.37	86.21	83.01
2		91.26	91.02	91.59	89.35	89.90	90.37	91.30	92.06	**92.44**
3		65.41	70.09	62.80	72.21	68.54	67.72	68.56	**72.25**	56.06
4		61.41	61.23	56.88	59.80	60.45	60.90	62.90	**67.87**	61.12
5		76.58	**83.03**	81.41	80.99	81.73	82.09	81.59	81.63	81.99
6		75.80	73.59	78.67	78.64	75.80	77.95	76.15	74.02	**81.98**
7		**67.73**	66.41	60.23	63.26	65.51	62.00	64.85	63.76	50.87
8		66.93	65.05	64.26	65.99	71.34	67.11	**72.04**	71.31	65.77
OA(%)		75.32	76.01	74.81	76.48	76.69	76.30	77.30	**77.68**	74.70
AA(%)		73.63	74.67	72.48	74.59	75.23	74.43	75.72	**76.14**	71.66
KAPPA(%)		71.16	72.02	70.53	72.53	72.80	72.31	73.50	**73.95**	70.35

The best results are marked in **bold**, and the second-best results are underlined.

MMSAF(Global), MMSAF(Local), and Normal, as shown in Fig. 9. Here, Global indicates global guidance, Local indicates local feature guidance, and Normal represents conventional multiscale aggregation. Experimental results demonstrate that MMSAF(Global) consistently outperforms MMSAF(Local) and Normal across all evaluation metrics on the three datasets, suggesting that global information guidance leads to more consistent performance improvements than local stacking, as shown in Table VII. In contrast, local stacking exhibits unstable results; for instance, on the GF-5 dataset, the SSIM of Local drops from 0.9942 to 0.9897, and ERGAS increases from 2.28 to 3.61. This indicates that solely stacking local features may introduce redundancy or overfitting, further validating the necessity of incorporating global dependencies.

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TABLE XIV
COMPARISON OF MODEL COMPLEXITY ACROSS DIFFERENT METHODS

Method	HDNET	AWAN	HSCNN	MST++	MSFN	RepCPSI	FESSN
Parameter(M)	11.46	21.53	<u>14.43</u>	26.72	61.01	15.96	**9.99**
Flop(G)	46.85	88.04	59.09	**24.66**	50.18	65.34	<u>26.67</u>
Time/ms	3.73	3.93	**2.22**	<u>2.38</u>	10.04	3.68	3.45

The best results are marked in **bold**, and the second-best results are <u>underlined</u>

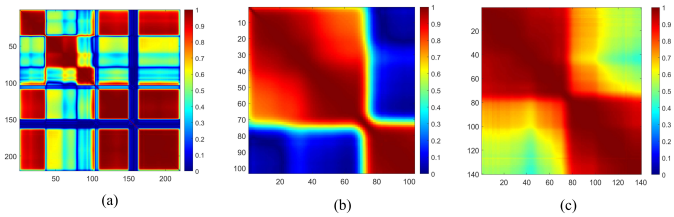

Fig. 11. Band correlation map. (a) IP dataset. (b) PU dataset. (c) GF-5 dataset.

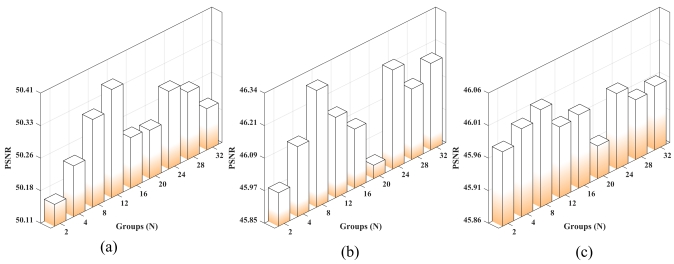

Fig. 12. Parameter sensitivity analysis of N. (a) IP dataset. (b) PU dataset. (c) GF-5 dataset.

B. Module Hyperparameter Analysis

1) Analysis of the Spectral Mamba Module: Our analysis indicates that after incorporating the Spectral Mamba module for global spectral information weighting, the RMSE on the IP, PU, and GF-5 datasets decreased by approximately 1.46%, 0.47%, and 0.72%, respectively. As shown in Table VIII and Fig. 11, the IP dataset exhibits more pronounced long-range and higher linear dependencies among spectral bands, allowing the Spectral Mamba module to more fully exploit global spectral correlations in this dataset. In contrast, the PU dataset contains a large number of distant, uncorrelated bands, which limits the performance gains from modeling long-range dependencies. For the GF-5 dataset, there are two distinct spectral ranges—the visible and near-infrared bands—that are largely separable; however, the abundant vegetation present maintains certain correlations between these bands, resulting in noticeable performance improvements. Collectively, these observations demonstrate that the global dependency modeling of the Spectral Mamba module is particularly effective for HSI reconstruction when significant long-range spectral correlations exist.

To evaluate the impact of the number of spectral groups on model performance, we set the group number N within the range {2, 4, 8, 12, 16, 20, 24, 28, 32} and measured their corresponding PSNR values. As shown in Fig. 12, all datasets exhibit performance degradation with fewer groups due to sparse weighting coefficients, which lead to missing interspectral dependencies. Based on the experimental results, we recommend using 8 as the optimal group number.

2) Analysis of Frequency-Enhanced Modulation: As shown in Table IX, in the simple scenario of the IP dataset, the combined Phase and Amplitude (P+A) modulation achieves the best performance across all evaluation metrics, indicating that the complementary information between phase and amplitude can significantly enhance reconstruction quality. This observation corroborates the classical theory in the Fourier domain, where phase primarily governs structural integrity while amplitude determines energy distribution; their synergy allows a more comprehensive preservation of both spectral and spatial features. In more complex scenarios, like the PU and GF-5 datasets, amplitude-only (Amp) modulation demonstrates superior performance in a particular index. For instance, in the PU dataset, it achieves an RMSE of 0.004477. This suggests that amplitude information contributes more directly to pixel-level numerical accuracy, and in the presence of noise or outliers, amplitude modulation can be more robust by neglecting high-frequency disturbances contained in the phase. Overall, amplitude governs signal energy distribution and plays a decisive role in numerical precision. Meanwhile, phase preserves structural correlations, leading to improvements in visual fidelity metrics such as PSNR and SSIM across all three datasets. The combined modulation of phase and amplitude thus achieves stronger robustness and fidelity.

C. Similarity Evaluation on Real Datasets

To further verify the practicality and robustness of the proposed method under real-world imaging conditions, we conducted experiments on real satellite data. The real Sentinel-2 MSI were acquired contemporaneously with the GF-5 observations and were preprocessed to obtain analysis-ready data under the same band settings and training conditions as in Section IV-B. As shown in Table X, on the real dataset, deeper networks with larger parameter sizes, such as MST++ and MSFN, gain certain advantages compared with their performance on simulated data. In contrast, smaller models like HSCNN and AWAN exhibit inferior error metrics. Benefiting from their compact yet effective architectural designs, RepCPSI and our proposed model exhibit stronger feature extraction capability and achieve superior reconstruction performance while maintaining low parameter complexity. This also demonstrates that our method retains high robustness when facing real-world degradations such as SRF mismatch, spatial blurring, sensor noise, stray light, and field of view misregistration.

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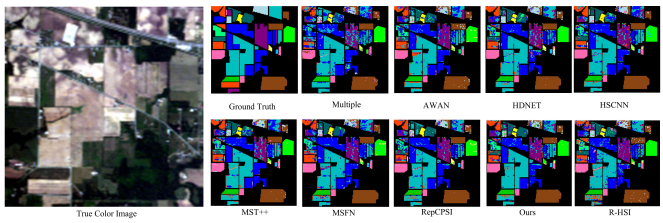

Fig. 13. Visual result comparison on the IP dataset.

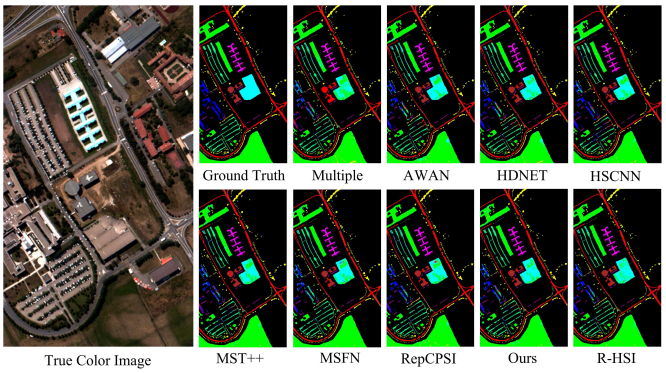

Fig. 14. Visual quality comparison on the PU dataset.

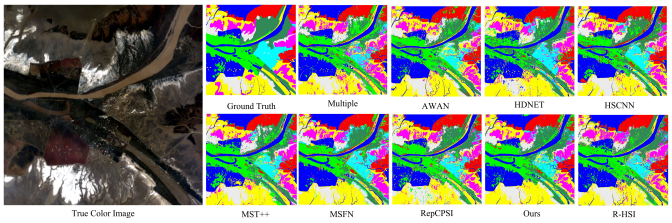

Fig. 15. Visual quality comparison on the GF-5 dataset.

D. Classification Evaluation

To evaluate the practical applicability of different reconstruction results in downstream tasks, we conducted classification experiments on three representative datasets: IP, PU, and GF-5, which differ in class number, distribution, and sample size. The IP dataset contains 16 land-cover classes with complex and highly mixed distributions, posing significant challenges for classification. The GF-5 dataset includes eight classes with many labeled samples, but relatively small spectral differences, leading to modest performance. In contrast, the PU dataset has fewer classes with high interclass separability, yielding stable and superior results even with limited samples.

For all experiments, we employed the classical support vector machine (SVM) as the classifier and performed five random training–testing splits to mitigate potential biases caused by sample partitioning and to ensure fair, reproducible evaluation. In addition to commonly used quantitative metrics, including OA, AA, and the Kappa coefficient (κ), we also reported per-class accuracies to provide a more detailed and fine-grained assessment of each reconstruction method's performance across different land-cover types. As shown in Tables XI–XIII and Figs. 13–15, the classification performance of HSI reconstructed by our method consistently approaches that of the real HSI across all metrics. Notably, our method achieves the highest number of best per-class accuracies as well as a substantial number of second-best results across the three datasets. These findings demonstrate that our approach not only effectively preserves the spectral–spatial characteristics of the original HSI but also exhibits strong robustness and generalization capability in complex, real-world scenarios.

E. Computational Cost

We evaluated the computational characteristics of each model using an input configuration of (1, 13, 64, 64) with 220 output spectral bands. The comparison, detailed in Table XIV, reveals the computational tradeoffs for each method. While HSCNN is the fastest at 2.22 ms, and MST++ has the lowest FLOPs at 24.66 G, our proposed FESSN demonstrates a superior overall balance. FESSN is the most lightweight model, achieving the lowest parameter count at 9.99 M. It also delivers the second-best FLOPs at 26.67 G and maintains a highly competitive inference speed of 3.45 ms. This highlights FESSN's exceptional design, which balances model compactness with high computational efficiency.

VI. Conclusion

In this study, we present the FESSN, a lightweight and high-performance neural architecture for reconstructing HSI from MSI acquired by satellite and airborne platforms. FESSN pioneers a multidomain fusion paradigm—integrating spatial, spectral, and frequency cues—while prioritizing computational efficiency with only 9.99 M parameters. At its core, the MMSAF module harnesses global contextual guidance from Mamba to inform depth-wise separable convolutions across multiple scales (3×3, 5×5, 7×7 kernels), enabling robust multiscale spatial feature extraction. Complementing this, the USEM module, built on a U-Net backbone, incorporates the SEM to capture intricate spectral dependencies: attention mechanisms handle local intragroup correlations, while Mamba models nonlinear intergroup global interactions. Furthermore, the FEM innovatively employs FFT with learnable linear layers to adaptively modulate spectral amplitudes and phases, mitigating local optima in spatial–spectral representations and enhancing overall fidelity. Extensive experiments on the IP, PU, and GF-5 datasets reveal FESSN's superiority over six state-of-the-art baselines, with notable gains in key metrics such as RMSE (up to 5.31% improvement), PSNR, SAM, ERGAS, and SSIM. Downstream evaluations in land-cover classification further affirm its practical efficacy, underscoring FESSN's role as a transformative advance in efficient, interpretable, and high-fidelity HSI reconstruction for remote sensing applications.

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